

PRACTIAL 3

AIM: Make a Snake game in pygame.

CODE:

```
import pygame
import time
import random
pygame.init()
width = 600
height = 400
window = pygame.display.set_mode((width, height))
pygame.display.set_caption("Snake Game")
white = (255, 255, 255)
black = (0, 0, 0)
red = (213, 50, 80)
green = (0, 255, 0)

clock = pygame.time.Clock()
snake_block = 10 # size of snake segment
snake_speed = 9 # speed of snake
def draw_snake(snake_block, snake_list):
    for x in snake_list:
        pygame.draw.rect(window, green, [x[0], x[1], snake_block, snake_block])
def game_loop():
    game_over = False
    game_close = False

    x = width / 2
    y = height / 2

    x_change = 0
    y_change = 0

    snake_list = []
    snake_length = 1

    food_x = round(random.randrange(0, width - snake_block) / 10.0) * 10.0
    food_y = round(random.randrange(0, height - snake_block) / 10.0) * 10.0
    while not game_over:

        while game_close:
            window.fill(white)
```

```

font = pygame.font.SysFont(None, 50)
message = font.render("You Lost! Press Q-Quit or C-Play Again", True, red)
window.blit(message, [width/6, height/6])
pygame.display.update()

for event in pygame.event.get():
    if event.type == pygame.KEYDOWN:
        if event.key == pygame.K_q:
            game_over = True
            game_close = False
        if event.key == pygame.K_c:
            game_loop()

for event in pygame.event.get():
    if event.type == pygame.QUIT:
        game_over = True
    if event.type == pygame.KEYDOWN:
        if event.key == pygame.K_LEFT:
            x_change = -snake_block
            y_change = 0
        elif event.key == pygame.K_RIGHT:
            x_change = snake_block
            y_change = 0
        elif event.key == pygame.K_UP:
            y_change = -snake_block
            x_change = 0
        elif event.key == pygame.K_DOWN:
            y_change = snake_block
            x_change = 0
    x += x_change
    y += y_change

if x >= width or x < 0 or y >= height or y < 0:
    game_close = True
window.fill(white)
pygame.draw.rect(window, red, [food_x, food_y, snake_block, snake_block])

snake_head = []
snake_head.append(x)
snake_head.append(y)
snake_list.append(snake_head)

if len(snake_list) > snake_length:

```

```

del snake_list[0]

# Snake collides with itself
for segment in snake_list[:-1]:
    if segment == snake_head:
        game_close = True

draw_snake(snake_block, snake_list)

pygame.display.update()
if x == food_x and y == food_y:
    food_x = round(random.randrange(0, width - snake_block) / 10.0) * 10.0
    food_y = round(random.randrange(0, height - snake_block) / 10.0) * 10.0
    snake_length += 1
clock.tick(snake_speed)
pygame.quit()
quit()

game_loop()

```

OUTPUT:

Output Video: Click here to watch

☐ [Click here to watch the output video](#)

Output Image:

